The Economic Impact and Market Forces of Shellfish Harvesting

November Project Update

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Gulf of Maine Research Institute

Introduction and Aims

Increase understanding of landings, ex-vessel prices, and additional factors impacting Maine soft-shell clam fisheries and markets

Topic 1: Understanding values and interactions associated with the soft-shell clam

Topic 2: Factors that affect soft-shell clams, shellfish price structures and pricing fluctuations

Goals of Project

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Provide Economic Analysis and Information to Support a Better Understanding of Soft-Shell Clam Landing Price Trends and Fluctuations at the Municipal, County, Regional, and State Levels

Analysis of Data: FAO Fish Price Index

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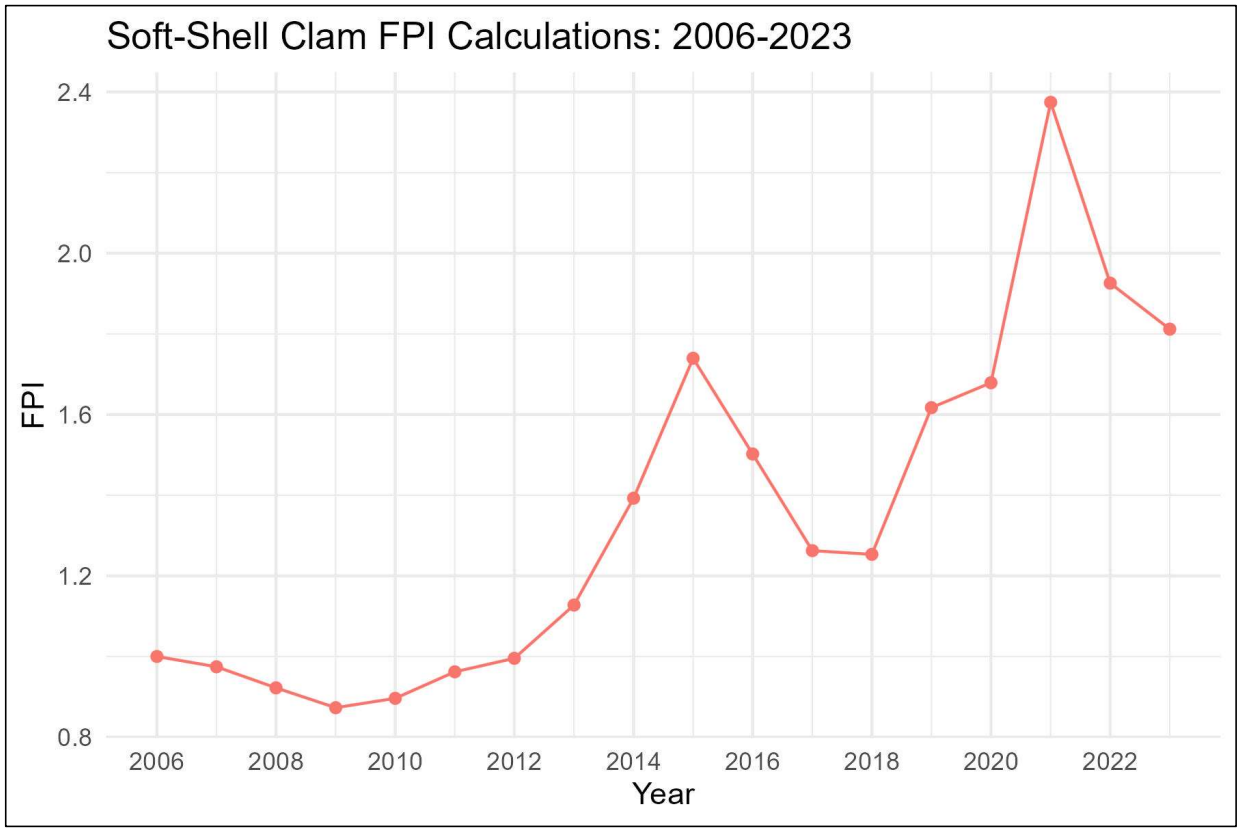
- FAO's Fish Price Index was developed to track changes in global or regional seafood availability (Tveterås et al. 2012)

- We will use this index to examine overall trends and changes in the ex-vessel prices of commercially important bivalve species (softshell clam, quahog, oyster) between 2006 and 2023



Soft-Shell Clam Prices by Region

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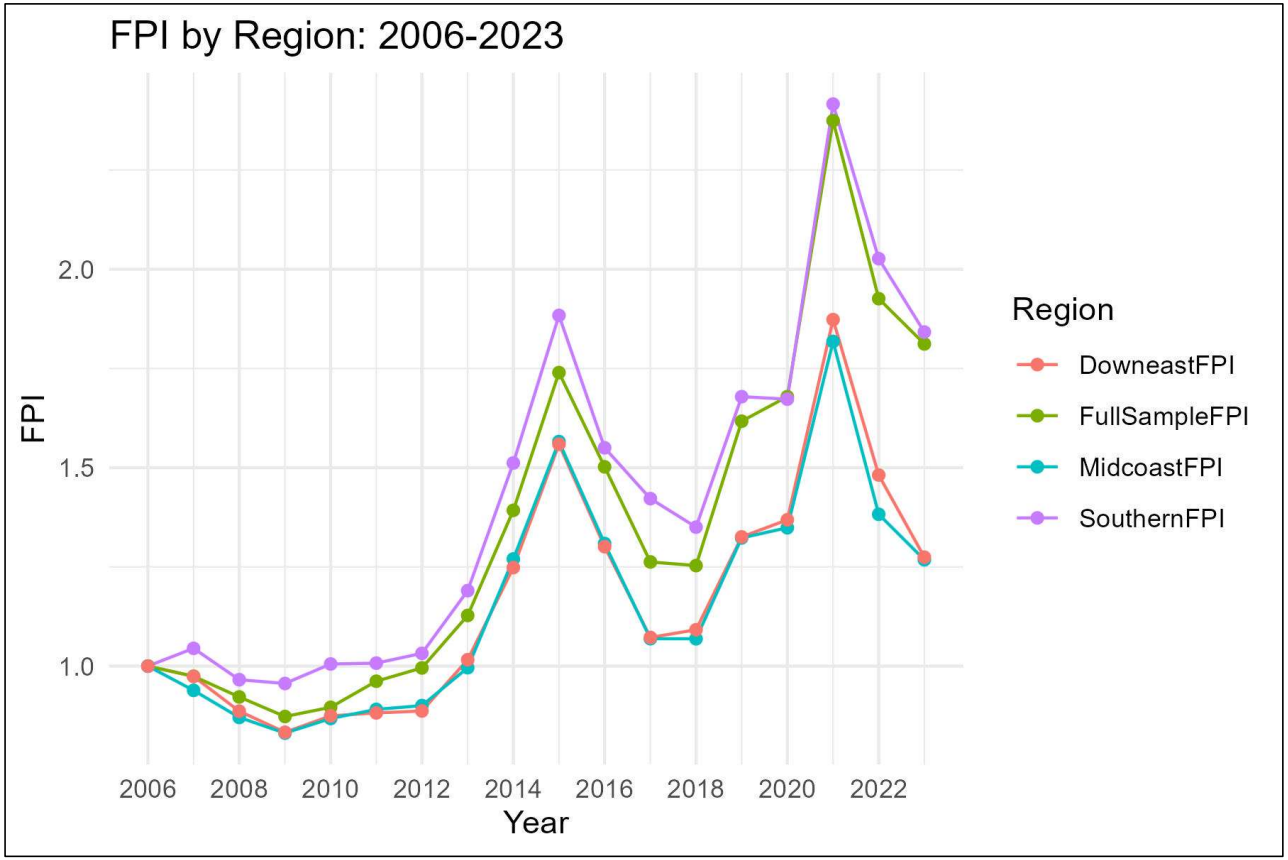
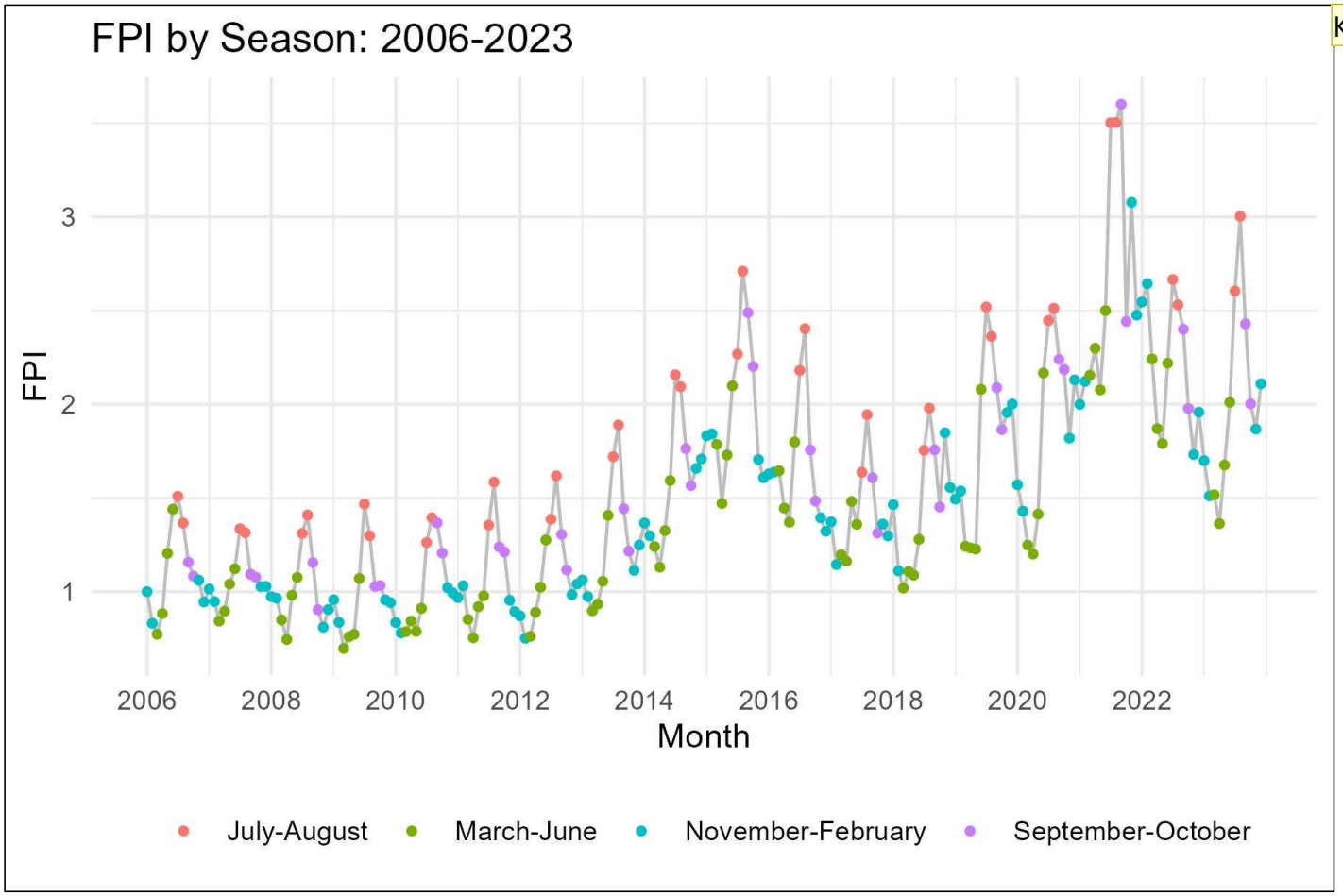
- Trends are consistent across three regions regarding price and production
- Southern Maine reports the highest price overall, gaps between Southern Maine and other two regions seemed to have widened in recent years
- Price peaked in 2021 and have been declining in recent years



Seasonal FPI



KT1

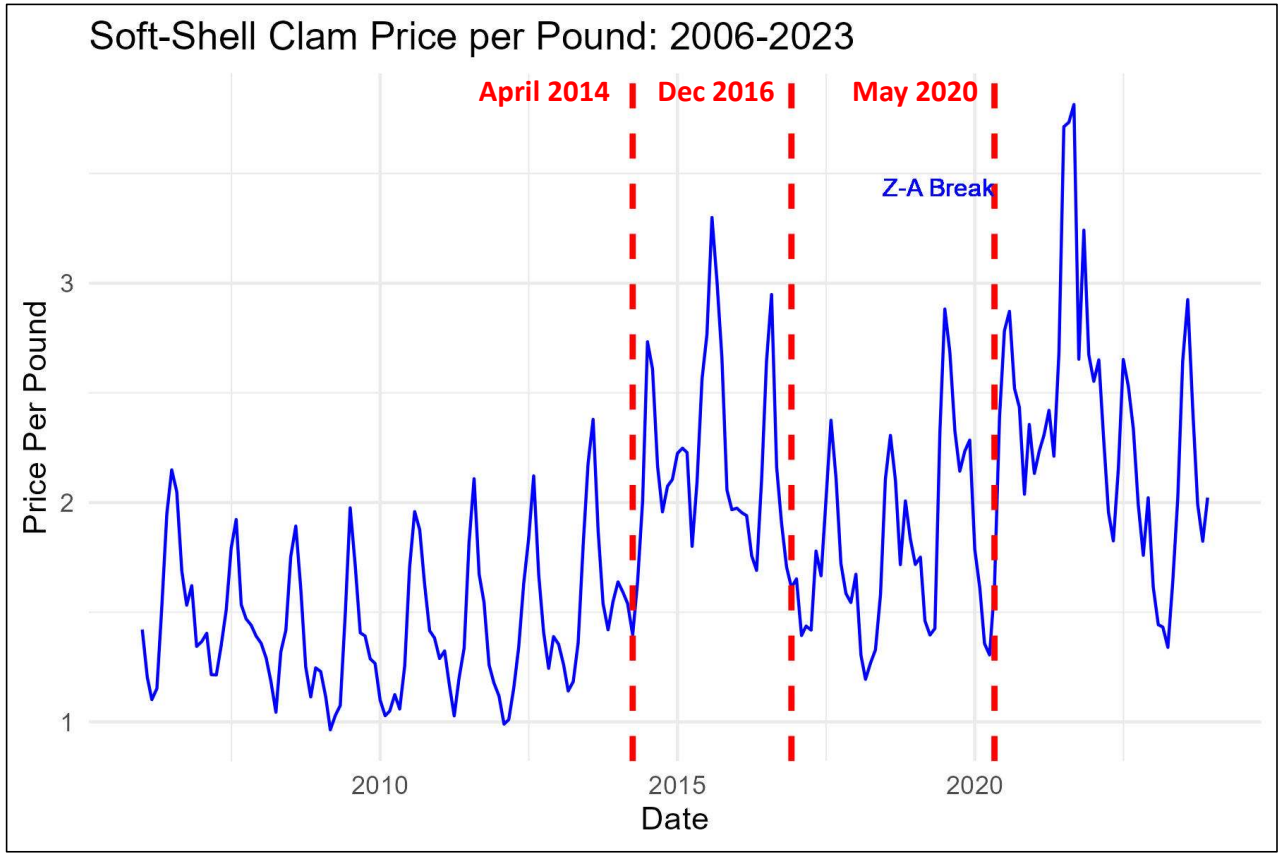
- Seasons were determined based on feedback from the last ShAC meeting
- We see consistently higher FPI indices during the July-August season
- Certain years depict significant changes that occur month to month

Slide 6

KT1 @bcotton@gmri.org There is something funny going on in this figure. I think you may be still pulling monthly FPI rather than seasonal. Can you double check and let me know if you'd like to discuss this in person.

Kanae Tokunaga, 2024-11-15T15:16:26.335

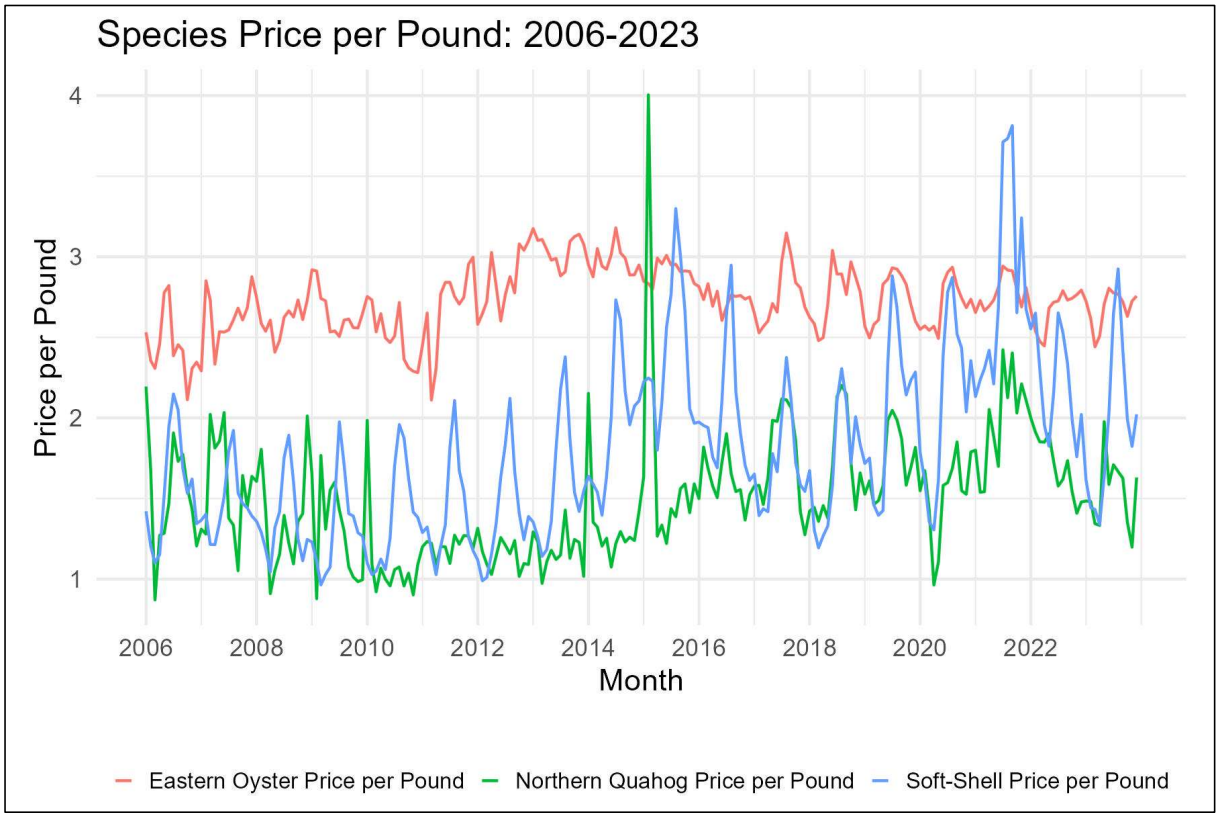
Shift in Landing and Price Trend



- Zivot-Andrews Tests were conducted on all species and across all regions to see any 'structural' shifts in landings and prices

- For Soft-Shell Clams, results suggest three trend breaks in April of 2014, December of 2016, and May of 2020 for landed Price per Pound

Market Integration



- Co-integration tests help to understand the market relationship between different regions (Southern, Mid-Coast, Downeast) and different species (Clam, Quahog, Oysters)

- Test results suggest there are long-term relationships between both regional and species average prices
  - Confirming that 3 regions are serving the same market and 3 species are serving the same market, potentially passing through the same supply chain channel

Next Steps + Additional Thoughts

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- Examine landings and ex-vessel price patterns to investigate environmental, macroeconomic, and other socioeconomic factors that influence them

- Predation (Green Crab, Ribbon Worm, etc.)
- Harmful Algal Blooms
- Pollution/Water quality
- Rainfall Closures
- Intertidal access
- COVID-19
- Ocean warming

→ Would like feedback on which factors are of the most interest